Buckling of Columns

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I. Introduction & Background

Objective: Understanding the phenomenon of column buckling, and measuring experimentally the buckling (or critical) load for columns under different boundary conditions.

A. (Q1) Definition of Buckling

Buckling is the sudden deformation of a structural member that is loaded in compression, that occurs when the compressive load in the member reaches a critical value.

B. (Q2) Difference Between Buckling and Bending

Bending occurs when a structural member is subjected to transverse load (the force acts perpendicular to the member's length). This causes the member to develop internal stresses, which in turn cause it to curve in a predictable and controlled manner. Buckling, on the other hand, refers to a sudden change in shape (deformation) and represents a state of instability in a structural member when it is subjected to a compressive load.

The difference between buckling and bending is as follows:-

1) In bending, deflection is in the same direction as the load, whereas in buckling, deformation is perpendicular to the direction of the load.
2) Bending is caused by transverse loads, whereas Buckling is caused by compressive loads.
3) Buckling occurs only after the compressive load reaches a critical value, but bending occurs for every value of load.
4) Deformation in case of bending occurs in the plane of loading, whereas in buckling, Deformation occurs perpendicular to the axis of compression.

C. (Q3) Can a slender body buckle without bending? Can it bend without buckling?

No, a slender body cannot buckle without bending, as buckling refers to the lateral deflections due to bending caused by compressible axial load under instability.
Yes, bending can occur without buckling (like a cantilever beam under transverse load), where the deflections are due to bending included by transverse loads under stable conditions.

D. (Q4) Dependence of Parameters

The second moment of Area (I), length of the column, and the effective length factor (K) which depends on the boundary conditions dictates the effective buckling length (KL) are all geometric properties.
The Young's Modulus (E) is the only material property in the formula for calculating (P_{cr}).

E. (Q5) Determining Radius

The formula for critical load (P) is given by:-

$$\boldsymbol{P_{cr} = \frac{\pi^2 EI}{(KL)^2}}$$

The second moment of area (I) for a circular cross-section of radius R is:

$$I = \frac{\pi R^4}{4}$$

For column of a circular cross-section of radius R and length L Initial critical load P_1:

$$P_1 = \frac{\pi^2 E I_1}{(KL_1)^2} = \frac{\pi^2 E\left(\frac{\pi R^4}{4}\right)}{(KL)^2}$$

For column of a circular cross-section of radius R' and length $2L$ Initial critical load P_2:

$$P_1 = \frac{\pi^2 E I_2}{(KL_2)^2} = \frac{\pi^2 E\left(\frac{\pi R'^4}{4}\right)}{(2KL)^2}$$

Equating the critical loads we get:-

$$\boxed{\boldsymbol{R' = \sqrt{2}R}}$$

II. Procedure

- Fix the column between the pin joints such that the initial load is zero and no buckling occurs initially.
- Calibrate the force-measuring device to zero.
- Incrementally apply load to the column and record the force reading on the digital device.
- Introduce a small perturbation by lightly tapping the table with a hammer.
- Record the new force reading on the digital device.
- Repeat the loading and perturbation steps until the column buckles, i.e., when increasing the load does not significantly change the force reading. At this stage, the change in reading after the perturbation remains constant.
- Repeat the above steps for the same column after flipping it.
- Perform the same procedure for the next four columns.

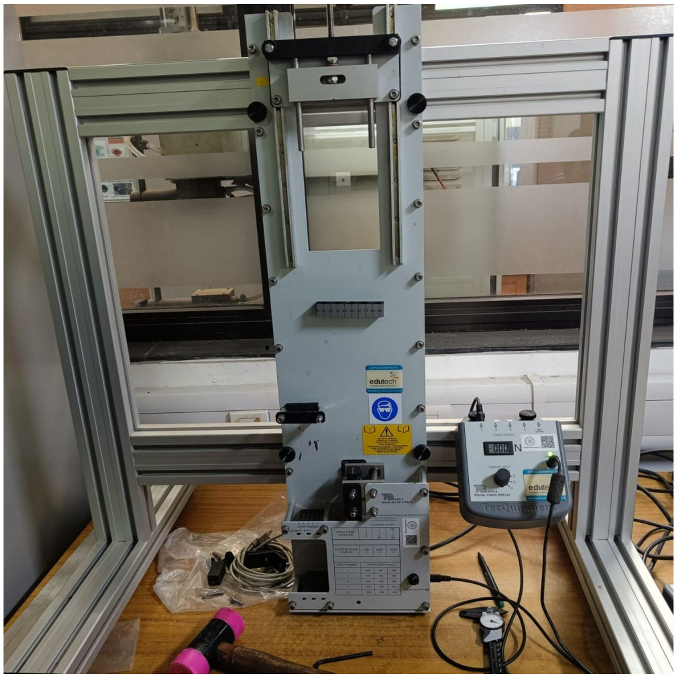

Fig. 1: Experimental Setup

III. Observations

In this section, we present the observed experimental values of the critical load for buckling of columns *in the first mode whose both ends are pinned* and calculate theoretical values for comparison.
The critical load for buckling is calculated for five different column lengths (L = 320 mm, 370 mm, 420 mm, 470 mm, and 520 mm) made of aluminium. For each given length of column, the reading was taken twice by reversing its orientation.

The theoretical critical load for buckling in the first mode with both ends pinned is given by:

$$\boxed{P_{\text{cr}} = \frac{\pi^2 EI}{L^2}} \tag{1}$$

Where E is the Young's modulus of the material, I is the second moment of area of the cross-section about the axis about which the column curves, and L is the length of the column. In this case,

- Young's Modulus (E): For aluminium, $E = 69$ GPa.
- Second Moment of Area (I): It is given by $I = bh^3/12$. Here, $b = 20$ mm, $h = 2$ mm. Therefore, $I = 13.33$ mm^4.

The experimentally determined values ($P_{\text{cr, exp}}$, $(P_{\text{cr, exp}})_{\text{avg}}$) and theoretically calculated values ($P_{\text{cr, theo}}$, using the above data and equation (1)) are summarised in the table as follows. The critical load versus column length plot is also shown in the figure 2. (**The link for the files containing raw experimental and calculated theoretical data is attached at the end**).

L (mm)	Orientation	$P_{\text{cr, exp}}$ (N)	$(P_{\text{cr, exp}})_{\text{avg}}$ (N)	$P_{\text{cr, theo}}$ (N)
320	1	69	68.5	88.56
320	2	68		
370	1	52	50	66.24
370	2	48		
420	1	34	34.5	51.41
420	2	34		
470	1	32	32	41.05
470	2	-		
520	1	13	14	33.54
520	2	15		

TABLE I: Experimental and Theoretical Critical Loads

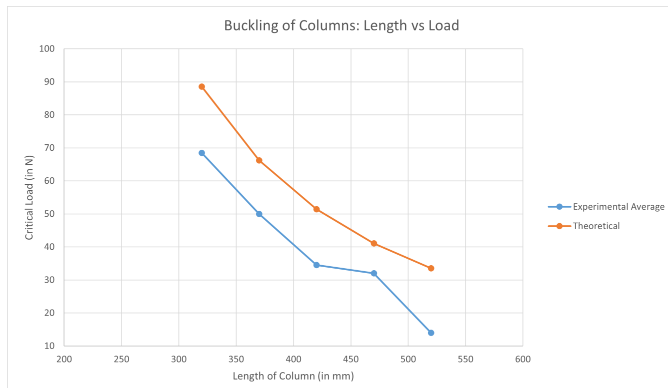

Fig. 2: Critical load (in N) versus length of column (in mm)

The error (in %) in the average experimental values with respect to the theoretical values is calcualted as Error = $|$average experimental value $-$ theoretical value$|$/theoretical value and is summarised in the table below.

L (mm)	$(P_{\text{cr, exp}})_{\text{avg}}$ (N)	$P_{\text{cr, theo}}$ (N)	Error (%)
320	68.5	88.56	22.65
370	50	66.24	24.52
420	34.5	51.41	32.89
470	32	41.05	22.05
520	14	33.54	58.25

TABLE II: Error between experimental and theoretical values

Observation and Interpretation: In theory, we observe that the critical load for buckling in the first mode of a column decreases with increasing length (as noted in equation (1), $P \propto 1/L^2$), or in other words, longer columns are prone to buckling as they have a small critical load.
Additionally, for much shorter columns, further decreasing the length significantly increases the critical load; conversely, for much longer columns, increasing the length by the same amount has a minimal effect on the critical load.
This trend is successfully observed in the experimentally observed values, the inverse relation between critical load and

length. However, a relatively constant offset is observed in Table II between theoretical and experimental values, likely indicating a systematic error, as shown in Figure 2. The possible reasons for the errors are discussed in the following sections.

IV. Sources of Error

- The self-weight of the column is not considered in the calculations, which can affect the measured critical load.
- The theoretical calculations use a standard value of Young's modulus (E), but the actual material may have a different or non-uniform E throughout the column.
- Misalignment of the column in the pin supports can introduce eccentric loading.
- Variation in the perturbation force: each hammer tap may apply a different force, leading to inconsistencies in the measured response of the column.

V. Conclusion

The experiment showed that the critical buckling load decreases as the column length increases, just as Euler's theory predicts. Our measurements followed the same inverse-square trend expected from theory. However, the experimental values were always slightly different from the theoretical values by almost a constant amount. This suggests that there were some systematic errors, possibly due to differences in the actual material properties or measurement inaccuracies. Even with these differences, the main buckling behaviour matched the theory well. This shows that the theoretical model is valid for our experiment within normal experimental limits.

Data Link

https://github.com/MukeshK17/Buckling-data.git

VI. References

1) "Understanding buckling," The Efficient Engineer, https://efficientengineer.com/buckling/ (accessed Nov. 15, 2025).